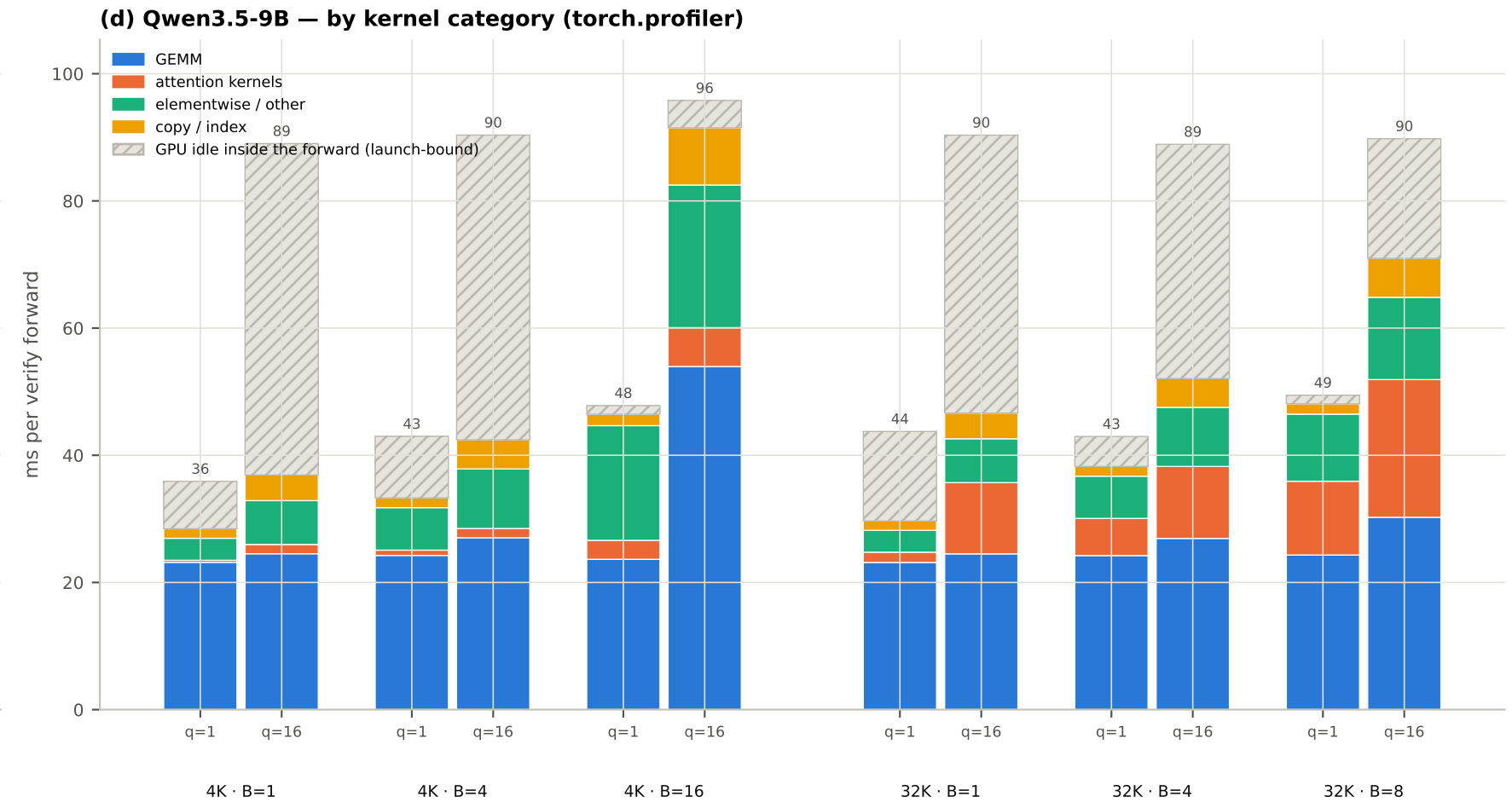
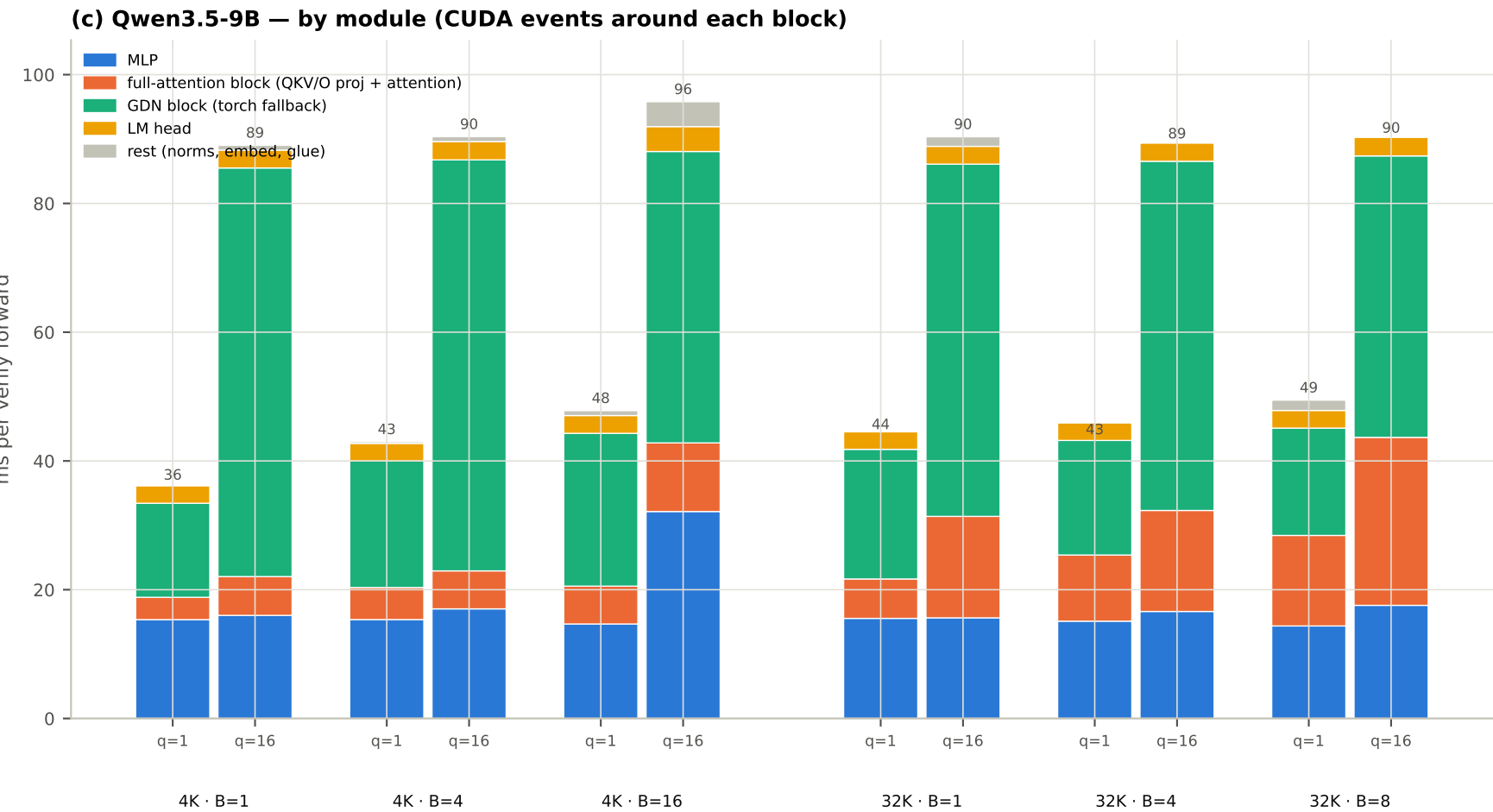
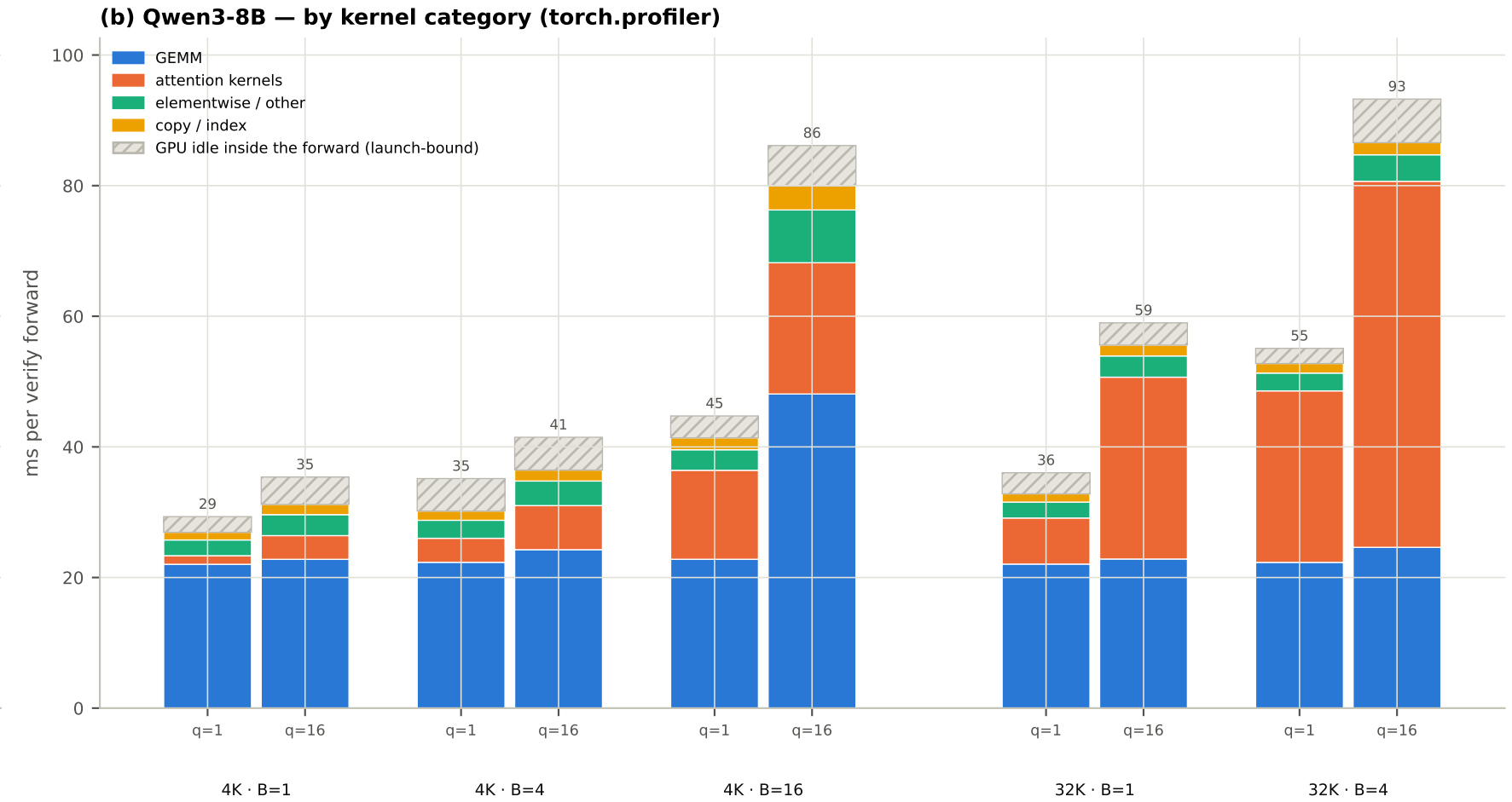
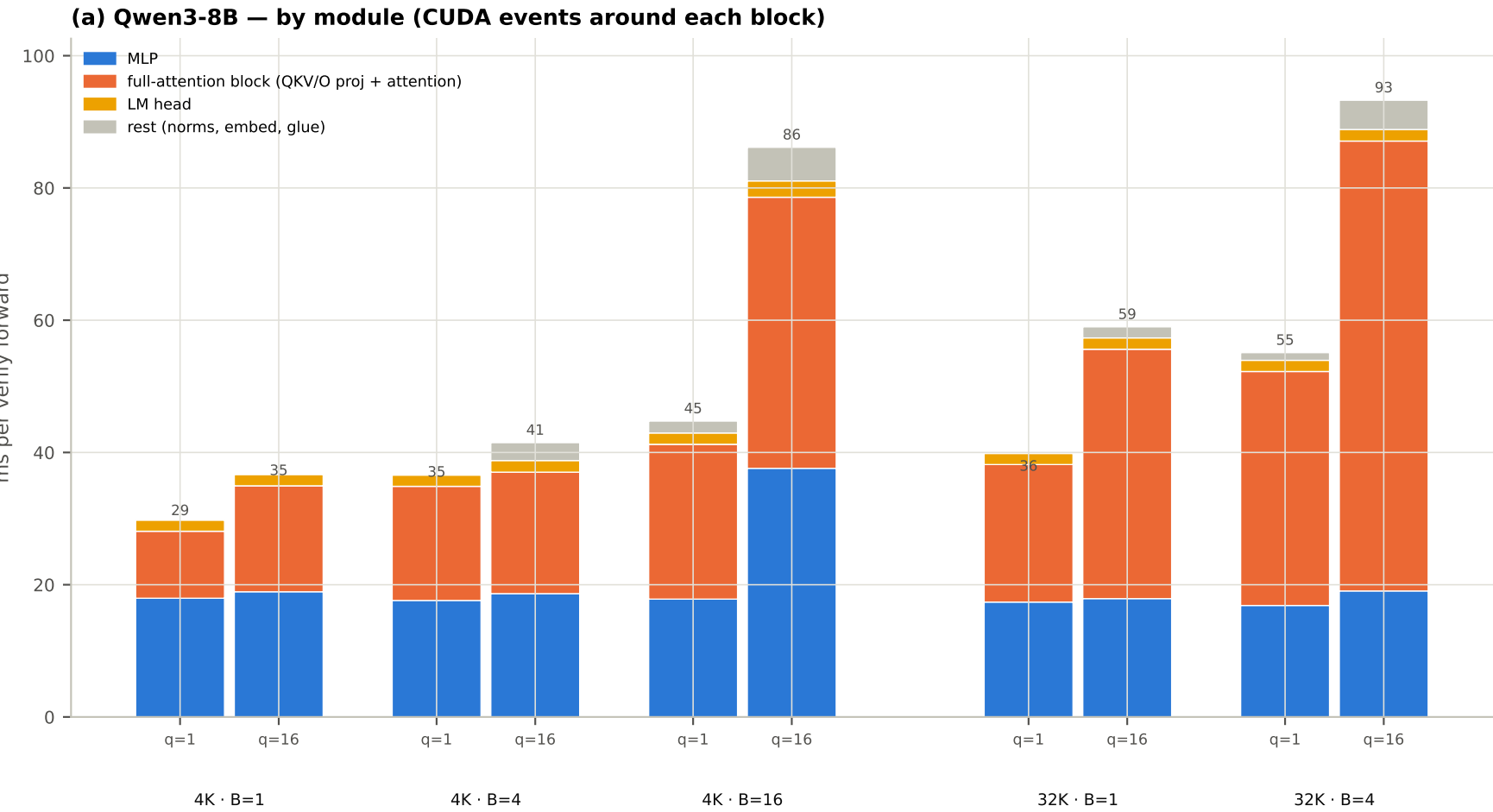


Verify forward, profiled at fixed shapes: q = 1 is the baseline's forward, q = 16 DFlash's



Measured outside the timed sweep (queue/verify_breakdown.py): one row prefilled and copied into the others, every forward rewriting the same slots. The hatched block is wall time the GPU spent without a kernel running — Python and launch overhead the kernels could not hide. For the 9B GDN torch fallback at q = 16 it is over half the forward; for 8B at 32K the attention kernel at q = 16 costs ~4× its q = 1 time at B = 1 because it does not share K/V across the query heads of a GQA group.